FAKULTI TEKNOLOGI DAN KEJURUTERAAN ELEKTRONIK DAN KOMPUTER

BERL3124 APPLICATION SYSTEM DEVELOPMENT 2
GROUP ASSIGNMENT PROJECT

Prepared by:

No.	Student Name	No. Matric	Picture
1.	**IRFAN BIN MAZUKI**	**B122310196**	
2.	**IEMRAN QAYYUM BIN ZULLEKA**	**B122310012**	
3.	**MUHAMMAD AZRUL BIN REDZUAN**	**B122310626**	
4.	**MOHAMMAD DANISH HAMIMI BIN NORHISHAM**	**B122310625**	

Prepared for:

EN. NOOR MOHD ARIFF BIN BRAHIN

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1.0 PROJECT CONCEPT AND CHOSEN DOMAIN

The Bus Ticket Management System is a web based application designed to manage and monitor bus ticket sales, customer information, and service performance through a centralized digital platform. The system handles ticket sales and customer records while also tracking the overall performance of bus services. This project aims to develop an automated solution that streamlines manual ticket handling processes, reduces human errors, and ensures that data is more accessible, accurate, and up to date.

The project focuses on the public transportation industry, specifically on bus ticket sales and passenger record management. As bus operators transport a large number of passengers daily, manual data recording methods are inefficient and prone to errors. By implementing this system, transportation companies can improve operational efficiency and make informed decisions through effective management of customer information and sales data analysis.

The system presents data to relevant personnel in visual formats such as bar charts, line graphs, and data tables. These visual tools support the analysis of ticket sales across different bus operators and enhance customer tracking, enabling better monitoring of service performance and business trends.

2.0 DESIGN DIAGRAMS

2.1 Use Case Diagram

3 actors:

i. *User*

ii. *Admin*

iii. *System*

8 use cases in total:

i. Create Account

ii. Search Bus Ticket

iii. Buy Ticket

iv. Pre-order Ticket

v. Login

vi. Manage Ticket

vii. View Purchased Ticket

viii. Process Payment

2 stereotypes:

i. One <<include>> relationship (mandatory sub-function).

- Login and Process payment

ii. One <<extend>> relationship (optional/conditional function).

- Pre-order Ticket

Actor	Use Case	Description
User	Create Account	User can create account in online shopping system.
User	Search Item	User can search item in online shopping system.
User	Buy Item	User can buy an item in online shopping system.
User	Pre-order Item	User can make pre-order for the item that not available yet
User	Login	User need to login if want to buy an item
Admin	Manage Item	Admin can manage item in online shopping system.
Admin	View Purchse Item	Admin can view purchase item in online shopping system.
System	Process Payment	System process payment when a user make a order.

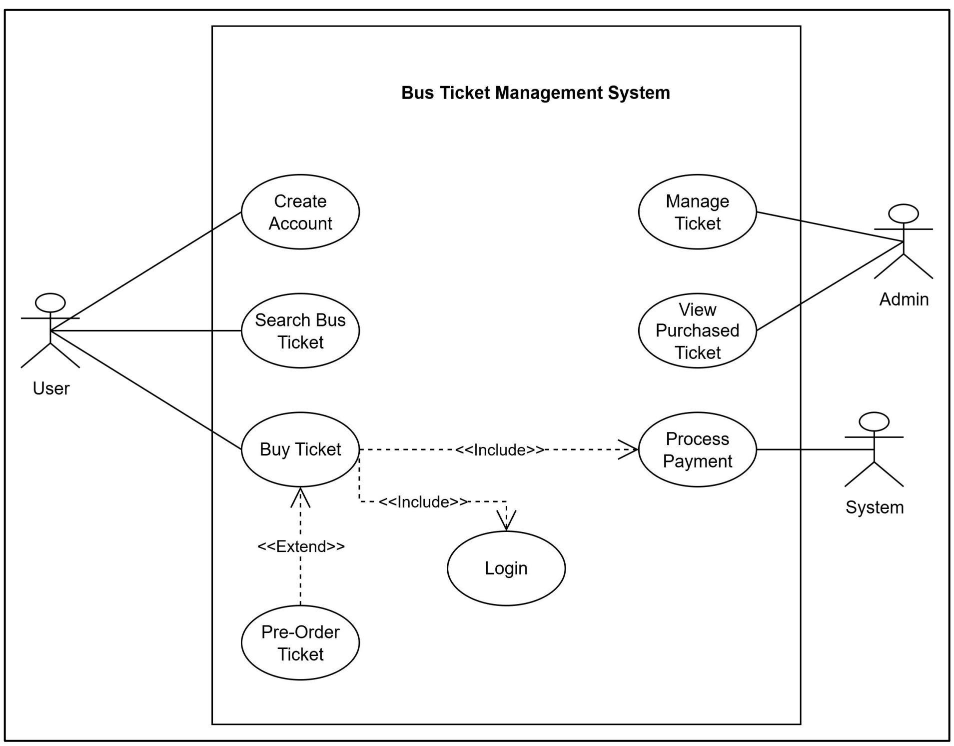
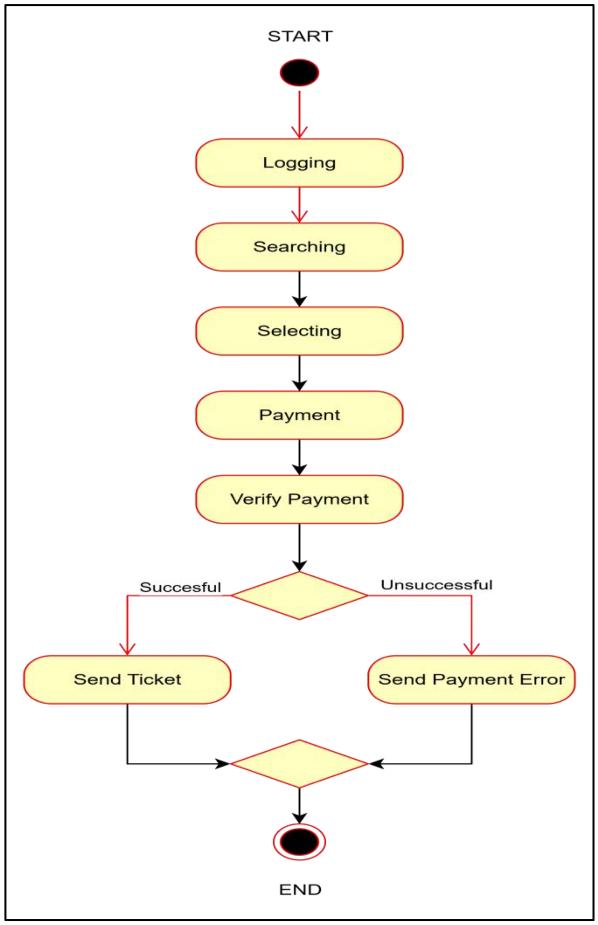


Figure2.1: Use Case Diagram

2.2 Activity Diagram



Fifure2.2: Activity Diagram

1. **Login** to the system.
2. **Search** for the bus ticket
3. **Select bus ticket** to add to the cart.
4. **Proceed to Checkout** once bus ticket are selected.
5. **Choose Payment Method**:
   - ➢ If the method is **valid**, proceed with **payment**.
   - ➢ If **invalid**, prompt the user to select **another method**.
6. Once payment is successful, **confirm order**.
7. **Log out** after the process completes.

2.3 Entity-Relationship Diagram

1. **Identify Entities and Attributes**

Entity	**Attributes**
Bus	Bus_ID (PK), Bus_name, Bus_direction, Ticket_price
Customer	Customer_ID (PK), Customer_Name
Ticket Buying	Bus_ID (PK) (FK), Customer_ID(PK) (FK)

2. **Determine Relationships and Cardinality**

Entities	**Entity Relationship**
Bus and Ticket Buying	One-to-Many
Customer and Ticket Buying	One-to-Many

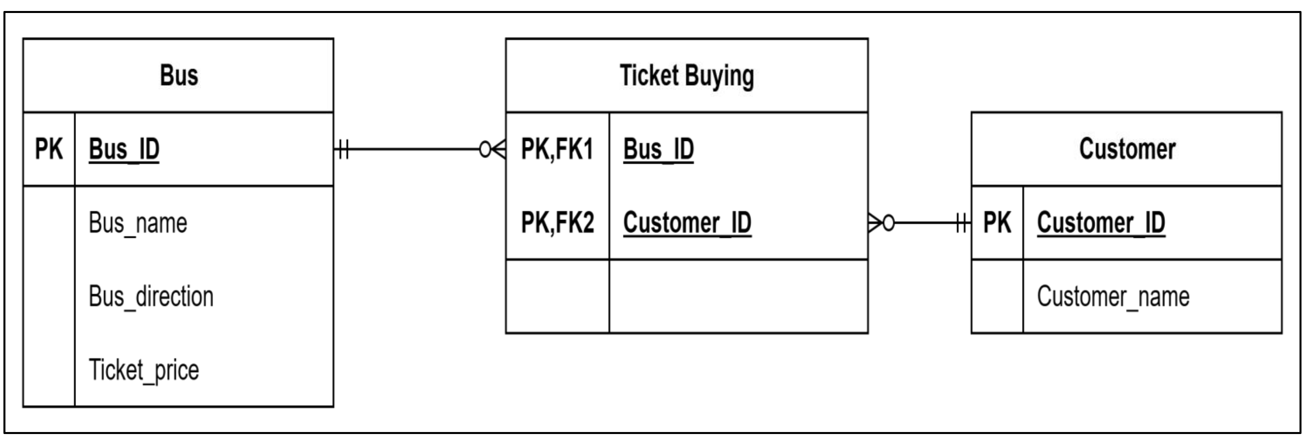


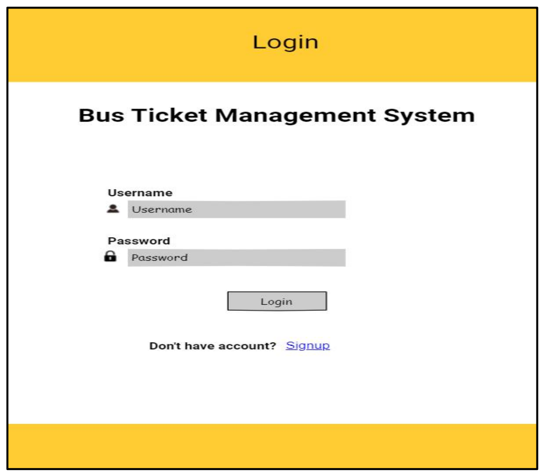
Figure2.3:Entity-Relationshiop Diagram

2.4 UI Wireframes Design

1. Wireframes

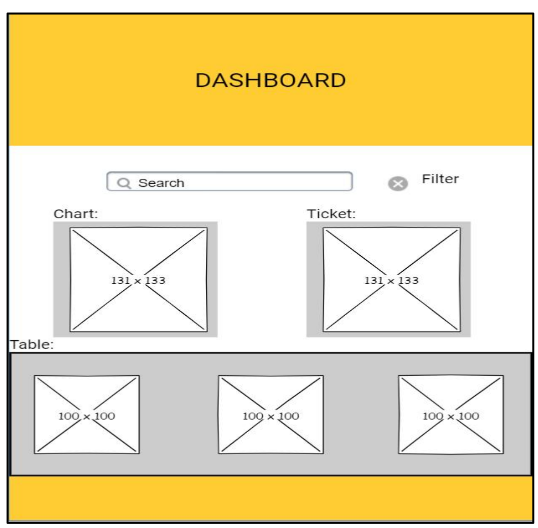
Screens

Login Page



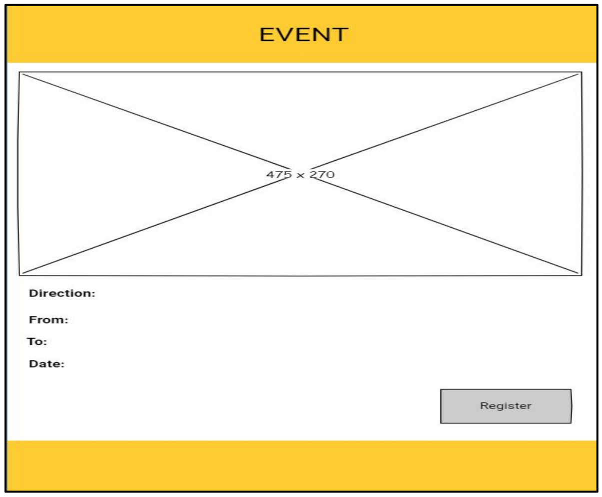
- ➢ Layout: Simple centered box with logo, username, password, and “Login” button.
- ➢ Links: “Signup” and “Forgot password?” under the form.

Dashboard Page



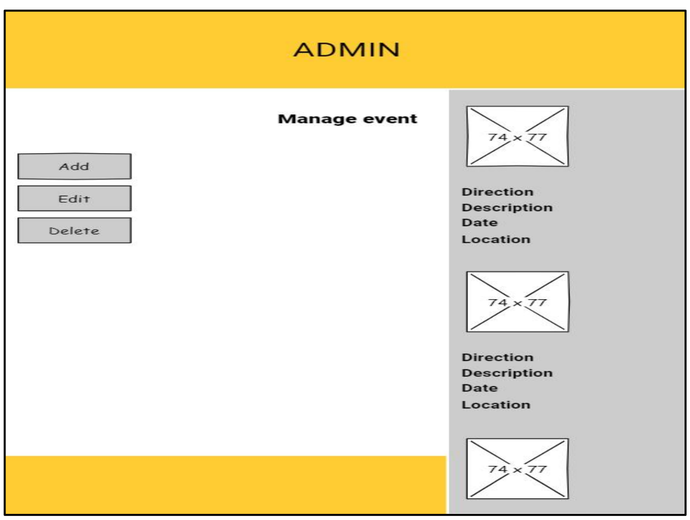
- ➢ Features: Search bar, and filter
- ➢ Navigation: Ticket → Event Details page.
- ➢ Accessibility: Clear text, good color contrast.

Event Details Page



- Shows full event info (direction, date, from, to).
- “Register” button for joining events.
- Accessibility: Easy to read, buttons are large and clear.

Admin Page



- Shows all events in a list with Edit/Delete buttons.
- “Create Event” button at the top.
- Includes form fields for event info.
- Accessibility: All fields labeled clearly.

3.0 SOURCE CODE

- FILE CODE getdata.php :

Edit File: getdata.php (PHP) Format

```php
<?php
// Database connection
$servername = "sql300.infinityfree.com"; // Replace with your MySQL hostname
$username = "if0_40493309"; // Replace with your MySQL username
$password = "Danishamimi1201"; // Replace with your MySQL password
$dbname = "if0_40493309_project_sir_arif"; // Replace with your MySQL database name

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT
    Bus.bus_name,
    COUNT(ticket_buying.customer_id) AS total_passenger
FROM Bus
LEFT JOIN ticket_buying
    ON Bus.bus_id = ticket_buying.bus_id
GROUP BY Bus.bus_id, Bus.bus_name;";
$result = $conn->query($sql);

$data = [];
while ($row = $result->fetch_assoc()) {
    $data[] = $row; //Appends the current row (associative array) to the $data array. This accumulate
}

echo json_encode($data);

$conn->close();
?>
```

This PHP script that connects to a database, queries bus names and passenger counts, and returns the data as JSON.

- FILE CODE data.php :

Edit File: data.php (PHP)

```php
// Database connection
$servername = "sql300.infinityfree.com"; // Replace with your MySQL hostname
$username = "if0_40493309"; // Replace with your MySQL username
$password = "Danishamimi1201"; // Replace with your MySQL password
$dbname = "if0_40493309_project_sir_arif"; // Replace with your MySQL database name

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Save form data
if (isset($_POST['submit'])) {
    $column1 = $_POST['column1'];
    $column2 = $_POST['column2'];


    $sql = "INSERT INTO Customer (Customer_id, Customer_name)
            VALUES ('$column1', '$column2')";

```

This PHP script that connects to the same database and handles form submissions by inserting new customer data into a table.

- FILE CODE index.php :

Edit File: index.php (PHP)

```
                                    <thead>
<tr>
    <th>Customer ID</th>
    <th>Customer Name</th>
</tr>
</thead>

       <tbody>
<?php
$servername = "sql300.infinityfree.com";
$username = "if0_40493309";
$password = "Danishamimi1201";
$dbname = "if0_40493309_project_sir_arif";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT Customer_id, Customer_name
FROM Customer
ORDER BY CAST(SUBSTRING(Customer_id, 2) AS UNSIGNED);";
$result = $conn->query($sql);

if ($result && $result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        echo "<tr>";
        echo "<td>" . $row["Customer_id"] . "</td>";
        echo "<td>" . $row["Customer_name"] . "</td>";
        echo "</tr>";
    }
} else {
    echo "<tr><td colspan='2'>0 results</td></tr>";
}

$conn->close();
?>
</tbody>
```

This PHP script that connects to a database and lists customer IDs and names in an HTML table.

- FILE CODE tables.php :

Edit File: tables.php (PHP)

```
                                        <thead>
                <tr>
                    <th>Customer Id</th>
                    <th>Customer Name</th>
                    <th>Bus Id</th>
                    <th>Bus Name</th>
                    <th>Bus Direction</th>
                    <th>Ticket Price</th>
                </tr>
            </thead>

            <tbody>
<?php
$servername = "sql300.infinityfree.com";
$username = "if0_40493309";
$password = "Danishamimi1201";
$dbname = "if0_40493309_project_sir_arif"; // sesuaikan dengan database kau

$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "
SELECT
    Customer.customer_id,
    Customer.customer_name,
    Bus.bus_id,
    Bus.bus_name,
    Bus.bus_direction,
    Bus.ticket_price
FROM Customer
LEFT JOIN ticket_buying
    ON Customer.customer_id = ticket_buying.customer_id
LEFT JOIN Bus
    ON ticket_buying.bus_id = Bus.bus_id
ORDER BY CAST(SUBSTRING(Customer.customer_id, 2) AS UNSIGNED);

";

$result = $conn->query($sql);

if ($result && $result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        echo "<tr>";
        echo "<td>" . $row['customer_id'] . "</td>";
        echo "<td>" . $row['customer_name'] . "</td>";
        echo "<td>" . $row['bus_id'] . "</td>";
        echo "<td>" . $row['bus_name'] . "</td>";
        echo "<td>" . $row['bus_direction'] . "</td>";
        echo "<td>" . $row['ticket_price'] . "</td>";
        echo "</tr>";
    }
} else {
```

This PHP script that retrieves and displays customer, bus, and ticket details from multiple database tables using SQL joins.

FILE CODE chart-area-demo.js :

Edit File: chart-area-demo.js (JavaScript)

```
// Set new default font family and font color to mimic Bootstrap's default styling
Chart.defaults.global.defaultFontFamily = '-apple-system,system-ui,BlinkMacSystemFont,"Segoe UI",Roboto,"Helvetica Neue",Arial,sans-serif';
Chart.defaults.global.defaultFontColor = '#292b2c';

// Area Chart Example
fetch('getdata.php')
            .then(response => response.json())
            .then(data => {
                const labels = data.map(item => item.bus_name);
                const salesData = data.map(item => item.total_passenger);

                const ctx = document.getElementById('myAreaChart').getContext('2d');
                new Chart(ctx, {
                    type: 'bar', // Change to 'line', 'pie', etc., as needed
                    data: {
                        labels: labels,
                        datasets: [{
                            label: 'Sales',
                            data: salesData,
                             backgroundColor: 'rgba(2,117,216,0.5)',
                            borderColor: 'rgba(0, 0, 0, 1)',
                            borderWidth: 2
                        }]
                    },
                    options: {
                        responsive: true,
                        scales: {
                            yAxes: [{
                                ticks: {
                                    beginAtZero: true
                                }
                            }
                                    ]
                        }
                    }
                });
            });
```

This contains JavaScript code that fetches JSON data and creates a bar chart to visualize passenger totals per bus.

- FILE CODE chart-bar-demo.js :

Edit File: chart-bar-demo.js (JavaScript)

```
// Set new default font family and font color to mimic Bootstrap's default styling
Chart.defaults.global.defaultFontFamily = '-apple-system,system-ui,BlinkMacSystemFont,"Segoe UI",Roboto,"Helvetica Neue",Arial,sans-serif';
Chart.defaults.global.defaultFontColor = '#292b2c';

// Area Chart Example
fetch('getdata.php')
            .then(response => response.json())
            .then(data => {
                const labels = data.map(item => item.bus_name);
                const salesData = data.map(item => item.total_passenger);

                const ctx = document.getElementById('myBarChart').getContext('2d');
                new Chart(ctx, {
                    type: 'pie', // Change to 'line', 'pie', etc., as needed
                    data: {
                        labels: labels,
                        datasets: [{
                            label: 'Sales',
                            data: salesData,
                             backgroundColor: ['#007bff', '#dc3545', '#ffc107', '#28a745', '#6f42c1' ],
                            //borderColor: 'rgba(2,117,216,1)',
                            borderWidth: 1
                        }]
                    },
                    options: {
                        responsive: true,
                        scales: {
                            yAxes: [{
                                ticks: {
                                beginAtZero: true
                            }
                            }
                                    ]
                        }
                    }
                });
            });
```

This contains JavaScript code that fetches the same JSON data but creates a pie chart to display passenger distribution per bus.

4.0 KEY FEATURES AND SYSTEM FUNCTIONALITY

4.1 Dashboard and Data Visualization

The system includes a dashboard that displays graphical representations of ticket sales using charts. An area chart is used to show sales comparison among different bus operators, while a pie or bar chart illustrates the proportion of sales contributed by each company. These visual tools allow administrators to quickly understand trends and performance without analyzing raw data.

4.2 Customer Management

The system provides a customer management module displayed in a data table format. Each customer is assigned a unique customer ID along with their name. The table supports pagination, sorting, and search functionality, allowing users to efficiently locate and manage customer records. This feature is useful for handling a large number of passengers and maintaining organized data.

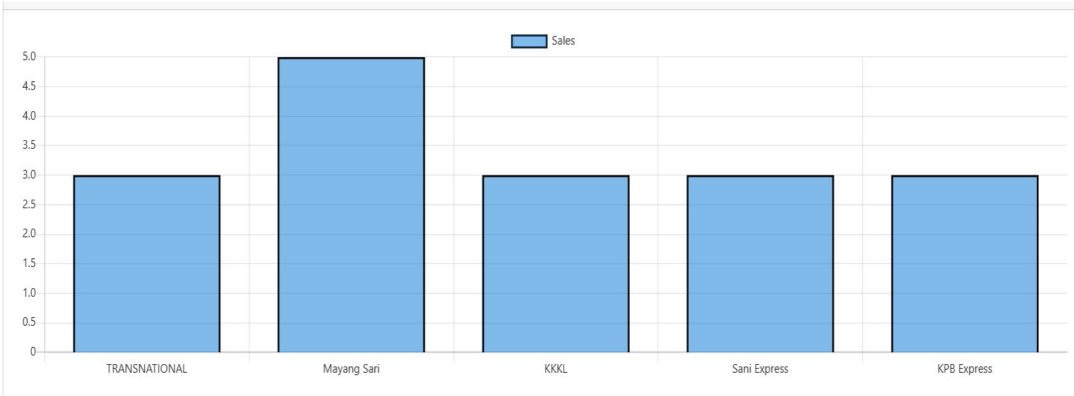
4.3 Sales Monitoring

The system tracks ticket sales for different bus operators such as Transnational, Mayang Sari, KKKL, Sani Express, and KPB Express. This functionality enables administrators to monitor sales performance and identify which operators are most frequently used by customers.

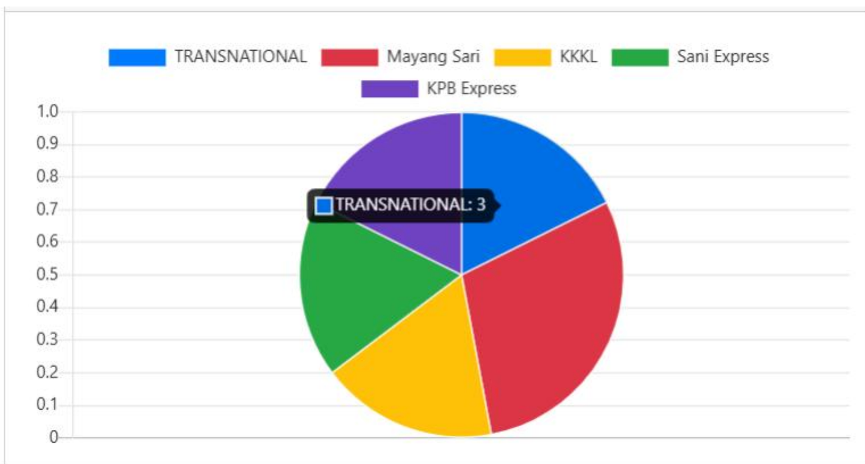
4.4 User Friendly Interface

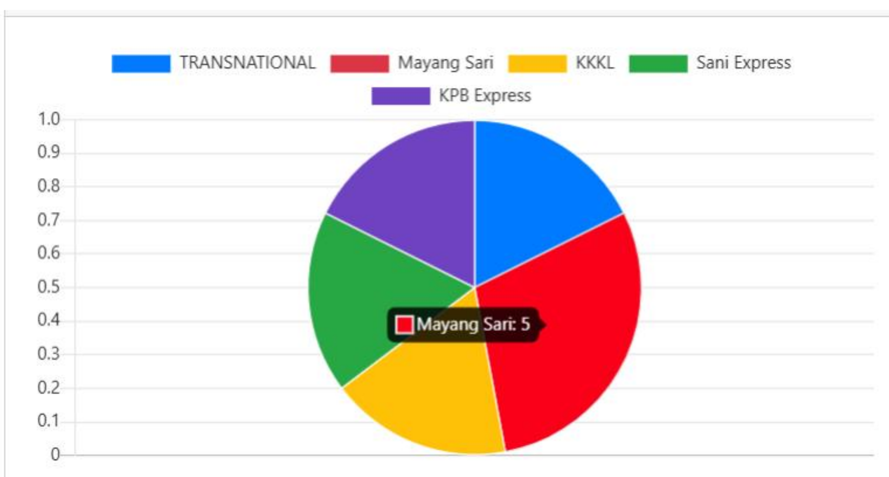
The interface is designed to be simple and easy to understand. Clear labels, structured tables, and visual charts help users interact with the system effectively, even without advanced technical knowledge.

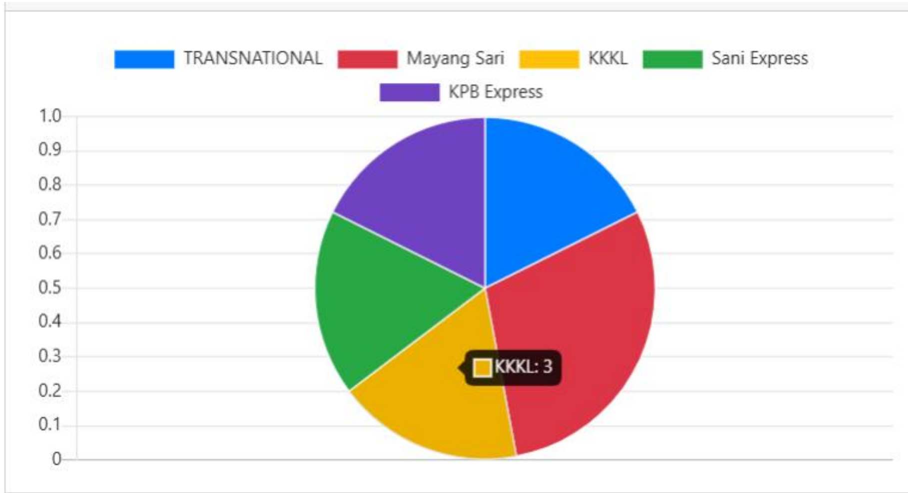
- BAR CHART :



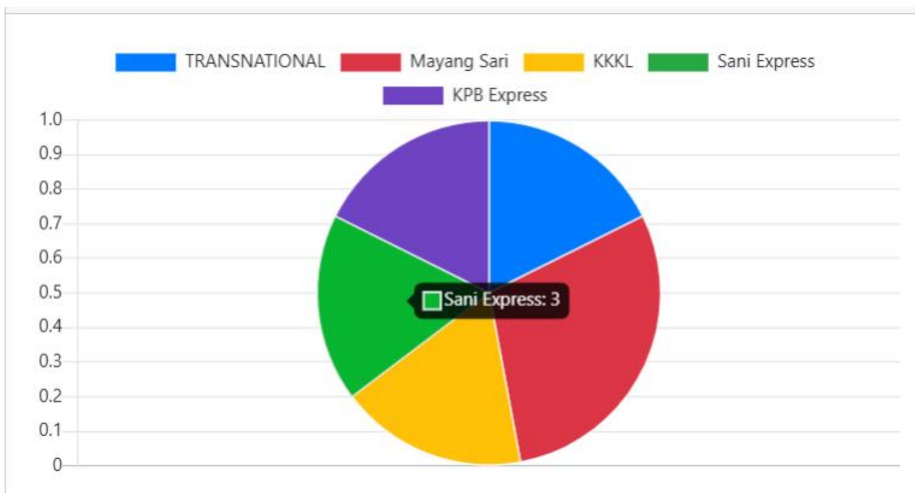
- PIE CHART :



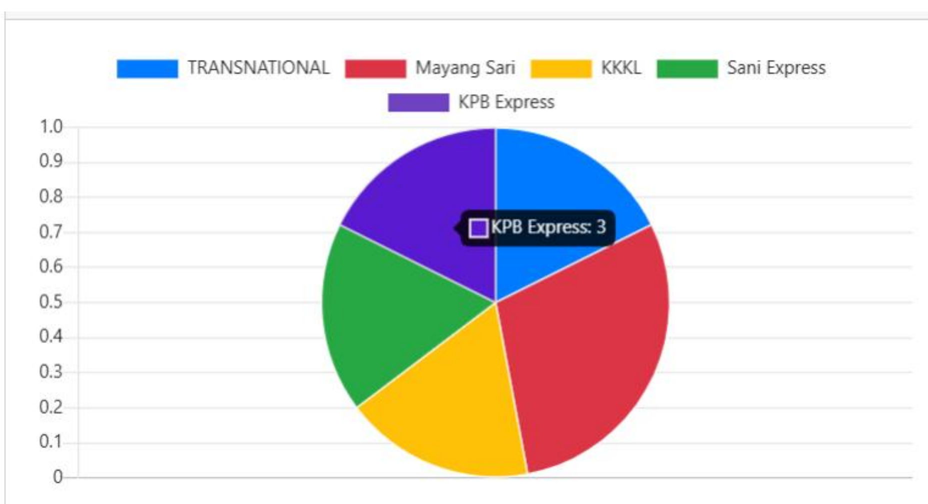




TRANSNATIONAL
Mayang Sari
KKKL
Sani Express
KPB Express
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
KKKL: 3



TRANSNATIONAL
Mayang Sari
KKKL
Sani Express
KPB Express
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Sani Express: 3



TRANSNATIONAL
Mayang Sari
KKKL
Sani Express
KPB Express
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
KPB Express: 3

DATA TABLE CUSTOMER :

DataTable Example

10 entries per page

Search...

Customer ID	Customer Name
C1	Danish Hamimi
C2	Mohd Azrul
C3	Ahmad Taufiq
C4	Qayyum
C5	Irfan
C6	Abdu Muaz
C7	Safuan
C8	Haziq
C9	Shahrul
C10	Nazriq

Showing 1 to 10 of 15 entries

‹ 1 2 ›

Customer ID	Customer Name
C11	Haris
C12	Faiz
C13	Faizal
C14	keco
C15	Loystopa

DATA TABLE CUSTOMER RECORD :

DataTable Example

10 entries per page

Search...

Customer Id	Customer Name	Bus Id	Bus Name	Bus Direction	Ticket Price
C8	Haziq	B5	KPB Express	PULAU PINANG - IPOH	RM55
C9	Shahrul	B1	TRANSNATIONAL	IPOH - KL	RM40
C9	Shahrul	B3	KKKL	MELAKA - KL	RM45
C10	Nazriq	B4	Sani Express	JOHOR - PERLIS	RM80
C11	Haris	B3	KKKL	MELAKA - KL	RM45
C12	Faiz	B2	Mayang Sari	MANJUNG - MELAKA	RM50
C13	Faizal	B1	TRANSNATIONAL	IPOH - KL	RM40
C14	keco				
C15	Loystopa				

Showing 11 to 19 of 19 entries

‹ 1 2 ›

DataTable Example

10 entries per page

Search...

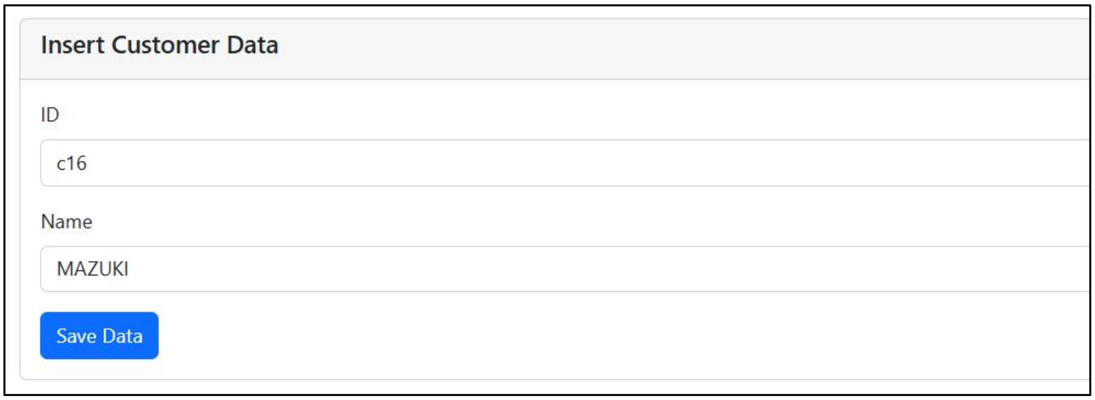
Customer Id	Customer Name	Bus Id	Bus Name	Bus Direction	Ticket Price
C1	Danish Hamimi	B5	KPB Express	PULAU PINANG - IPOH	RM55
C2	Mohd Azrul	B2	Mayang Sari	MANJUNG - MELAKA	RM50
C3	Ahmad Taufiq	B1	TRANSNATIONAL	IPOH - KL	RM40
C4	Qayyum	B2	Mayang Sari	MANJUNG - MELAKA	RM50
C4	Qayyum	B4	Sani Express	JOHOR - PERLIS	RM80
C5	Irfan	B2	Mayang Sari	MANJUNG - MELAKA	RM50
C5	Irfan	B5	KPB Express	PULAU PINANG - IPOH	RM55
C6	Abdu Muaz	B4	Sani Express	JOHOR - PERLIS	RM80
C7	Safuan	B3	KKKL	MELAKA - KL	RM45
C8	Haziq	B2	Mayang Sari	MANJUNG - MELAKA	RM50

Showing 1 to 10 of 19 entries

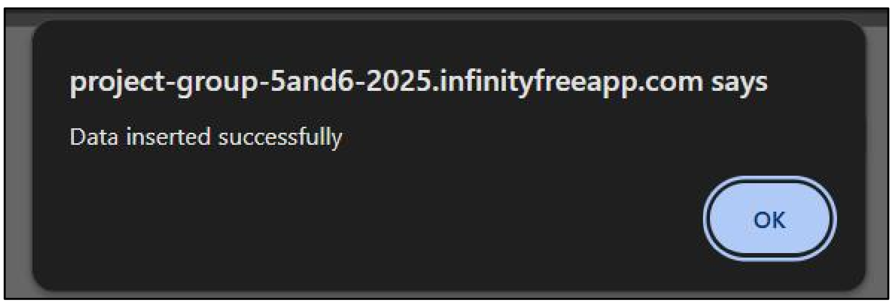
‹ 1 2 ›

- TABLE INSERT CUSTOMER DATA :

Step 1 :



Step 2 :



Step 3 :

5.0 FUTURE IMPROVEMENTS AND POTENTIAL EXPANSIONS

5.1 Online Ticket Booking

In the future, the system can be expanded to include an online ticket booking feature. This would allow customers to purchase tickets directly through the system, reducing the need for physical counters and improving convenience.

5.2 Payment Gateway Integration

The system can be enhanced by integrating online payment methods such as credit or debit cards and ewallets. This improvement would support cashless transactions and improve overall customer experience.

5.3 Seat Selection and Real Time Availability

A seat selection feature with real time seat availability can be added to allow customers to choose their preferred seats. This would reduce booking conflicts and improve service satisfaction.

5.4 Admin and Staff Role Management

Future versions of the system may include different user roles such as admin, staff, and manager. Each role can be assigned specific permissions to enhance security and operational control.

5.5 Reporting and Analytics

Advanced reporting features such as monthly sales reports, peak travel analysis, and customer behavior trends can be introduced. These analytics tools would help management make better strategic decisions.

6.0 CONCLUSION

In conclusion, the Bus Ticket Management System provides an efficient and organized solution for managing ticket sales and customer data in the bus transportation domain. With its visual dashboards, customer management features, and user friendly design, the system helps improve operational efficiency and data accuracy. With future enhancements such as online booking, payment integration, and advanced analytics, the system has strong potential to become a comprehensive digital solution for bus service providers.

7.0 LINK FOR WEBSITE AND PRESENTATION

1) **Link for InfinityFreeApp:**

project-group-5and6-2025.infinityfreeapp.com

2) **Link for presentation:**

https://youtu.be/RI23brqpTg4?si=CIX7xfxo9Cy0x0j3